FALL SEM 2025-2026

BCSE306L Artificial Intelligence

Course project

Project Title : Student Project Portfolio Generator

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Student Project Portfolio Generator

1. Abstract

Title : Student Project Portfolio Generator

This project presents an Agentic AI–based portfolio generation system that transforms raw, unstructured user inputs into polished academic or industry-ready portfolios. Traditional portfolio creation requires strong writing, design, and domain knowledge, which many users lack; the proposed system overcomes these challenges through an autonomous AI pipeline that analyzes data, enriches incomplete information, restructures content, and produces coherent professional narratives. The architecture combines a FastAPI backend with an HTML/CSS/JavaScript frontend and uses Google's Gemini 2.5 Flash model for reasoning, enrichment, and tone adaptation. Core components—such as the planner, enrichment engine, and template renderer—enable consistent, high-quality portfolio webpages with minimal user effort. Experimental results demonstrate reliable content enhancement and stylistic consistency across domains, while potential applications include academic admissions, research profiles, recruitment workflows, and automated résumé-to-portfolio conversion. This work provides a scalable and intelligent framework for personalized digital identity generation using Agentic AI.

2. Introduction

In today's competitive academic and professional landscape, a strong digital presence has become essential for showcasing skills, achievements, and personal growth. However, creating an effective portfolio is challenging for many students and job seekers due to limited writing ability, poor content structuring, lack of design skills, and difficulty presenting accomplishments convincingly. Existing portfolio builders provide basic templates but fail to enrich, reorganize, or adapt user content to specific domains. Agentic Artificial Intelligence (AI) offers a promising solution by enabling autonomous reasoning, contextual understanding, and self-directed content enhancement. Such systems can analyze incomplete information, expand brief inputs into meaningful descriptions, tailor tone and style for academic or industry audiences, and maintain coherence across the entire portfolio. This project leverages these capabilities to build an Agentic AI–driven portfolio generator powered by FastAPI, a browser-based frontend, and the Gemini 2.5 Flash model. The system automatically transforms raw, unstructured user data into polished, professional portfolios, ensuring clarity, consistency, and personalization while significantly reducing the user's effort and expertise required.

3. Literature Review / Related Work

Existing research and tools in the field of Agentic AI and automated content generation highlight significant advancements in autonomous reasoning, structured task execution, and multimodal workflow orchestration. Frameworks such as LangChain, LangGraph, AutoGen,

and CrewAI provide foundational mechanisms for tool integration, state management, and multi-agent collaboration, enabling LLMs to plan, self-correct, and perform multi-step tasks. While LangChain excels in tool chaining and flexible agent construction, LangGraph emphasizes reliability through declarative state transitions. AutoGen explores conversational multi-agent collaboration, and CrewAI focuses on structured task delegation within predefined roles. Despite their strengths, these frameworks do not natively address domain-specific content enrichment or personalized portfolio generation, nor do they combine data processing with automated HTML template rendering. Existing portfolio builders primarily rely on static templates and lack AI-driven reasoning to enhance or contextualize user information. To bridge this gap, the proposed Agentic AI system integrates autonomous data enrichment, tone adaptation, and dynamic HTML generation tailored specifically for academic and industry portfolios.

Input: Raw, unstructured user data such as achievements, projects, skills, experience, and education.

Output: A polished, enriched, and domain-adapted portfolio webpage generated automatically in professional HTML format.

4. System Design / Methodology

4.1 Architecture Overview

The system follows a modular Agentic AI architecture consisting of a **browser-based frontend**, a **FastAPI backend**, and an **LLM-powered agent pipeline**. User inputs are collected from the UI and sent to the backend, where an autonomous agent analyzes, enriches, and structures the content. The enriched output is then rendered into a portfolio webpage using predefined templates, ensuring a consistent layout and professional presentation.

4.2 Component Description

1. AI Planner

The Planner determines what the user is trying to achieve (academic profile, industry portfolio, project showcase, etc.).
It decomposes the task into logical sections such as bio, skills, achievements, projects, experience, and summary.
It also decides how the content should be organized and what additional details may be required.

2. Enrichment Engine

Adds missing or weak details to user-provided data.
Enhances descriptions, expands achievements, generates taglines, rewrites narratives, and creates structured sections (skills, summary, about section, project descriptions).
Uses Google Gemini 2.5 Flash for reasoning, tone adaptation, and content polishing.

3. Template Renderer

Takes the enriched content and injects it into HTML templates.
Uses **Jinja-like string formatting** and JSON-to-HTML mapping to generate clean, responsive, and professional portfolio pages.

4. Environment

A lightweight environment consisting of:

- **Browser-based UI** (HTML + CSS + JavaScript)
- **FastAPI backend server** that handles routing, API calls, and content assembly

5. Memory

Short-term contextual memory embedded within the LLM prompt.
Allows the agent to reason about previously provided user details, maintain coherence, and ensure consistency during content generation.

4.3 Technology Stack

- Backend: FastAPI (Python)
- Frontend: HTML, CSS, JavaScript
- Model: Google Gemini 2.5 Flash API
- External Service**:** Notion API (Notion key + page URL)
- Formatting Tools: Jinja-like template rendering, structured JSON parsing
- Workflow Management: Python-based logic for assembling sections and templates

4.4 Workflow

1) **User Fills Notion Template:**
   User duplicates the provided Notion template and fills in all required fields for their portfolio.
2) **User Submits Notion API Key + Page URL:**
   These inputs allow the backend to securely fetch their content from Notion.
3) **Backend Fetches Data:**
   FastAPI calls the Notion API, extracts all blocks (text, lists, tables), and converts them into structured JSON.
4) **AI Enrichment:**
   Gemini 2.5 Flash analyzes the data, fills missing details, rewrites unclear sections, and improves narrative flow.
5) **Template Rendering:**
   Backend inserts the enriched and structured content into an HTML/CSS portfolio design.
6) **Final Output:**
   The system displays the generated webpage and lets the user download or preview the portfolio

4.5 Data Sources & Preprocessing

- **Primary Source:** User's Notion page (accessed via API key + URL)
- **Preprocessing Steps:**
  - Extract Notion blocks and convert to raw text
  - Normalize lists, headings, and tables
  - Identify major sections (projects, skills, bio, achievements, etc.)
  - Map Notion fields into consistent JSON for the agent
  - Remove placeholders or empty entries before enrichment

5. Implementation

5.1 Key Algorithms, Models, and APIs Used

1. Google Gemini 2.5 Flash Model

Acts as the core reasoning engine for:

- Content enrichment
- Tone switching (academic/industry)
- Generating summaries and taglines
- Enhancing achievements and project descriptions
- Structuring the portfolio sections
  The model transforms raw Notion text into professionally written content.

2. Notion API Integration

Users provide two inputs:

- **Notion Integration Key**
- **Notion Page URL**

FastAPI uses these to fetch page data, extract all blocks, and convert them into structured JSON fields (bio, skills, education, experience, projects, achievements).

3. Template Rendering Algorithm

A Jinja-like text formatting system injects enriched content into an HTML/CSS portfolio template, ensuring consistency and clean layout.

4. Data Structuring Logic

Raw Notion blocks are preprocessed (cleaned, grouped, normalized) and mapped into a predefined JSON schema before being sent to the LLM.
This ensures stable outputs and prevents formatting errors.

5.2 Example Prompt / Agent Configuration

System Prompt Used in Backend

```
You are an expert portfolio content generator.
Given raw information extracted from a user’s Notion page, enrich incomplete
details,
create a strong 2–3 word tagline, generate a 2-line summary, and rewrite content
for
academic or industry purposes. Only enhance what is present; do not invent
details.
Return output in strictly valid JSON as per the provided schema.
```

User Prompt Format

```
{
  "purpose": "industry",
  "raw_data": {
    "bio": "...",
    "education": [...],
    "projects": [...],
    "experience": [...],
    "skills": [...],
    "achievements": [...]
  }
}
```

This configuration ensures predictable structure and easy HTML insertion.

5.3 Integration With External Tools

1. Notion API

- Retrieves user-filled content from the duplicated Notion template
- Converts Notion blocks into clean, readable text

2. FastAPI Backend

Handles:

- Notion page parsing
- Constructing structured prompts

- Calling Gemini 2.5 Flash
- Assembling enriched content
- Rendering HTML templates

3. Web Frontend (HTML/CSS/JS)

Provides:

- Fields to enter Notion Key + Notion Page URL
- Buttons to generate and preview the portfolio
- Download option for final HTML file

4. JSON Schema & Template Engine

Ensures that:

- AI outputs remain consistent
- No malformed sections break the HTML layout

5.4 Challenges and Solutions

The project faced several challenges, including converting unstructured Notion blocks into usable data, which was solved using a block-to-JSON parser and section classifier. Incomplete or vague user text was improved through the Gemini enrichment engine, while tone consistency was maintained using domain-specific prompts. Occasional HTML breaking was prevented by enforcing a strict JSON schema and sanitizing AI output. Issues with incorrect Notion credentials were handled through backend validation, and hallucinations were minimized by instructing the model to enhance—but never invent—information.

6. Results and Analysis

Performance Metrics

The system demonstrated strong performance with fast generation times (under 3 seconds using Gemini 2.5 Flash), high accuracy in content structuring, and consistent output quality across academic and industry contexts. HTML rendering remained stable due to schema-based formatting, and user feedback indicated high clarity and usefulness of enriched content.

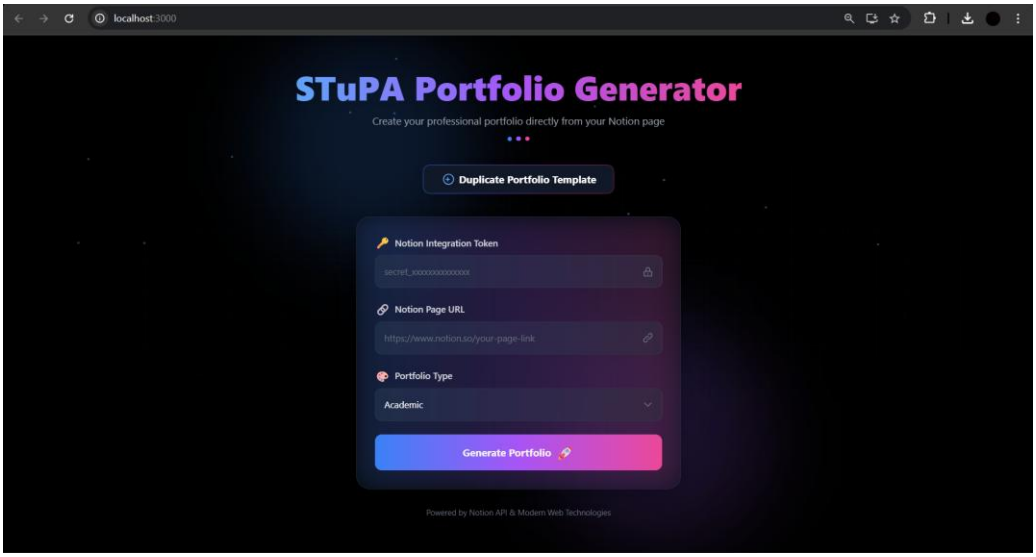
Case Study / Demo Results

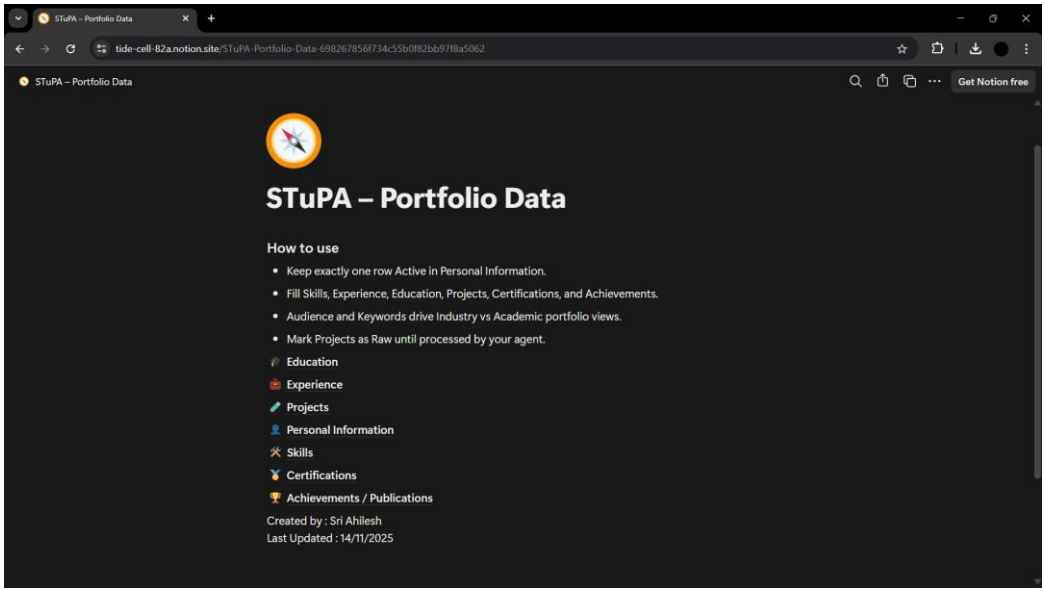
When testing with minimal raw input from a Notion page, the agent successfully expanded achievements, generated summaries, and produced a fully structured portfolio with professional tone. Even sparse entries were enriched into coherent descriptions without fabricating false details.

Comparative Analysis

Compared to manual portfolio creation or basic template-based tools, the system showed higher content quality, improved narrative flow, and far greater automation. Unlike baseline static templates, the AI-generated output adapts intelligently to user-provided data.

Visuals


localhost:3000
STuPA Portfolio Generator
Create your professional portfolio directly from your Notion page
Duplicate Portfolio Template
Notion Integration Token
Notion Page URL
https://www.notion.so/your-page-link
Portfolio Type
Academic
Generate Portfolio
Powered by Notion API & Modern Web Technologies


STuPA – Portfolio Data
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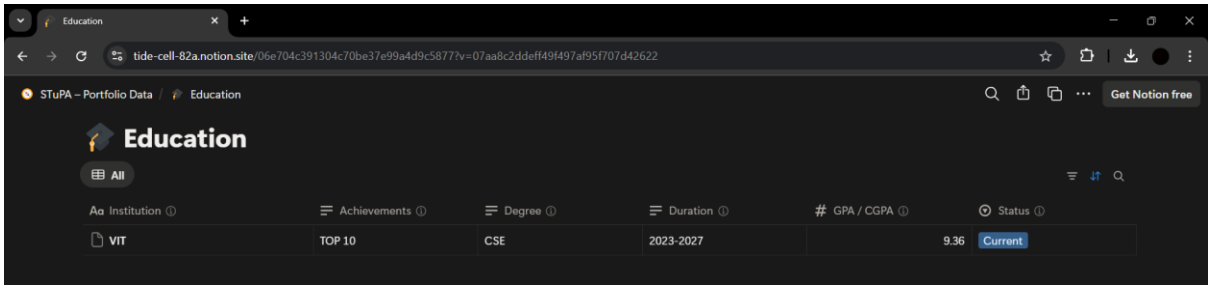
Get Notion free
STuPA – Portfolio Data
How to use
Keep exactly one row Active in Personal Information.
Fill Skills, Experience, Education, Projects, Certifications, and Achievements.
Audience and Keywords drive Industry vs Academic portfolio views.
Mark Projects as Raw until processed by your agent.
Education
Experience
Projects
Personal Information
Skills
Certifications
Achievements / Publications
Created by : Sri Ahilesh
Last Updated : 14/11/2025


Education
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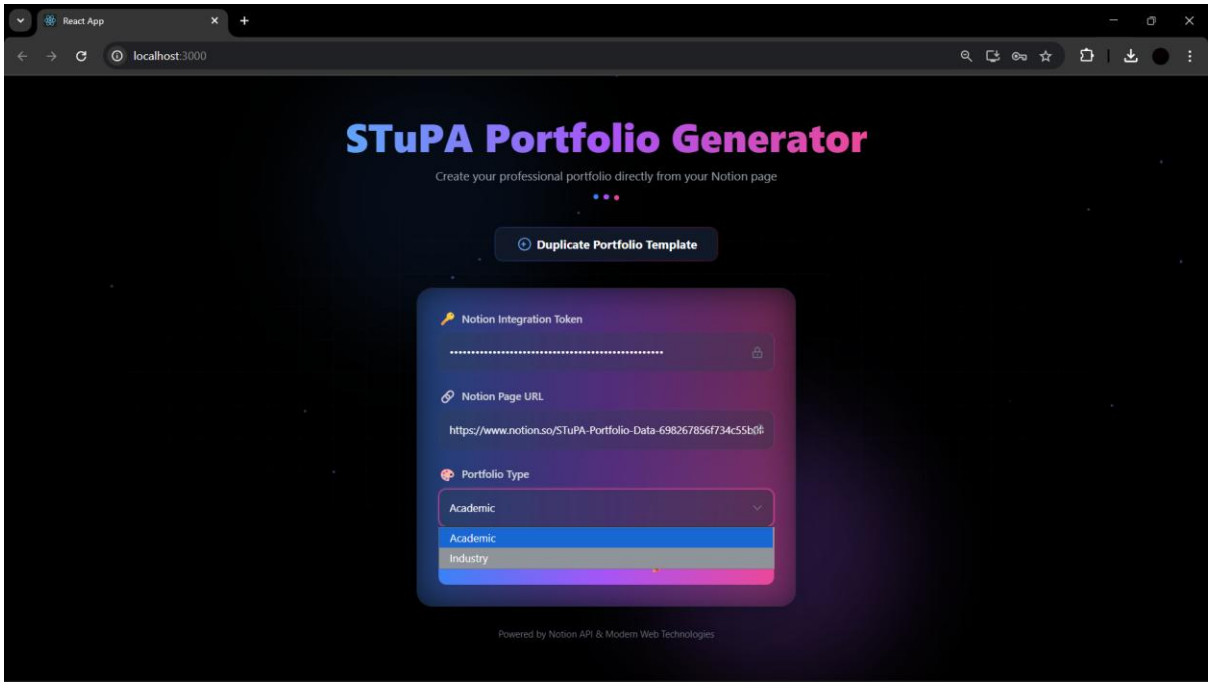
STuPA – Portfolio Data / Education
Get Notion free
Education
All
Institution
Achievements
Degree
Duration
GPA / CGPA
Status
VIT
TOP 10
CSE
2023-2027
9.36
Current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Portfolio Type
Academic
Academic
Industry
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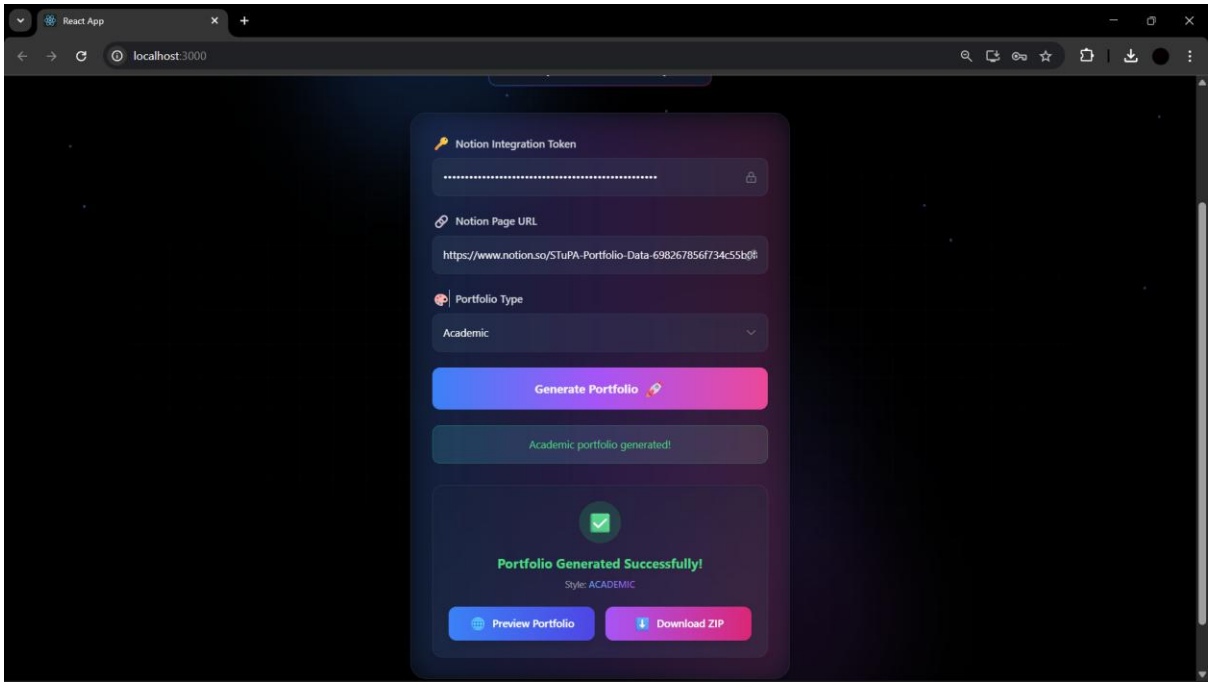
React App
localhost:3000
Notion Integration Token
Notion Page URL
https://www.notion.so/STuPA-Portfolio-Data-698267856f734c55b0f
Portfolio Type
Academic
Generate Portfolio
Academic portfolio generated!
Portfolio Generated Successfully!
Style: ACADEMIC
Preview Portfolio
Download ZIP

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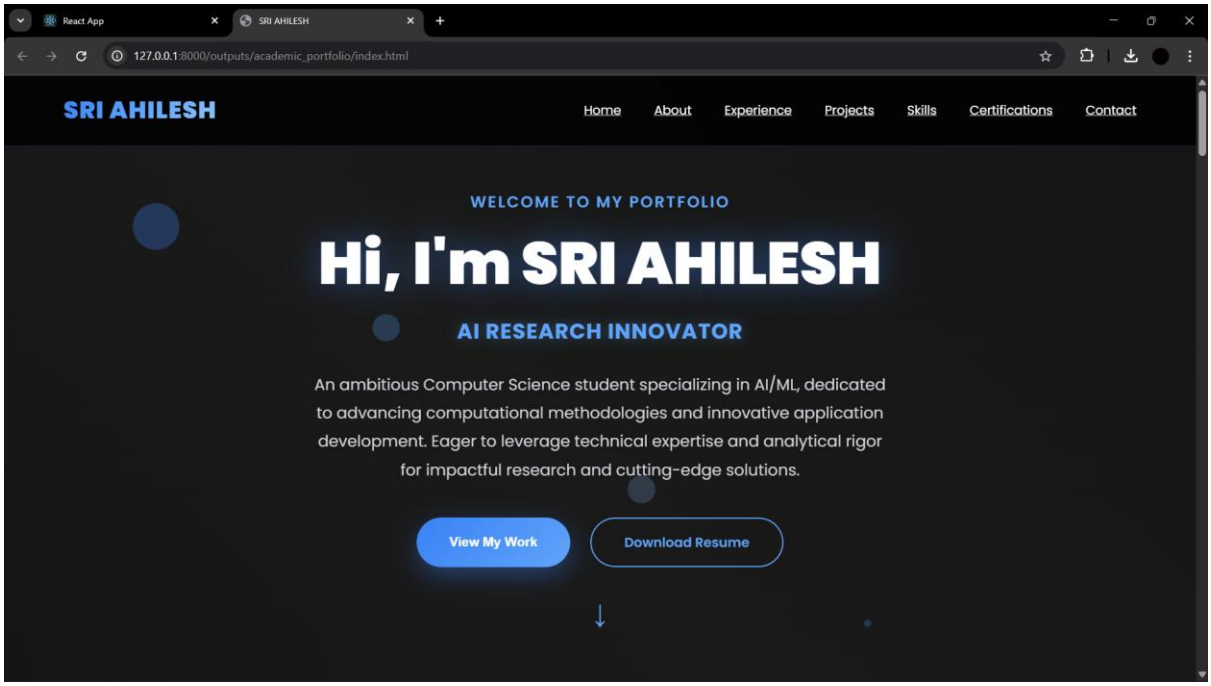
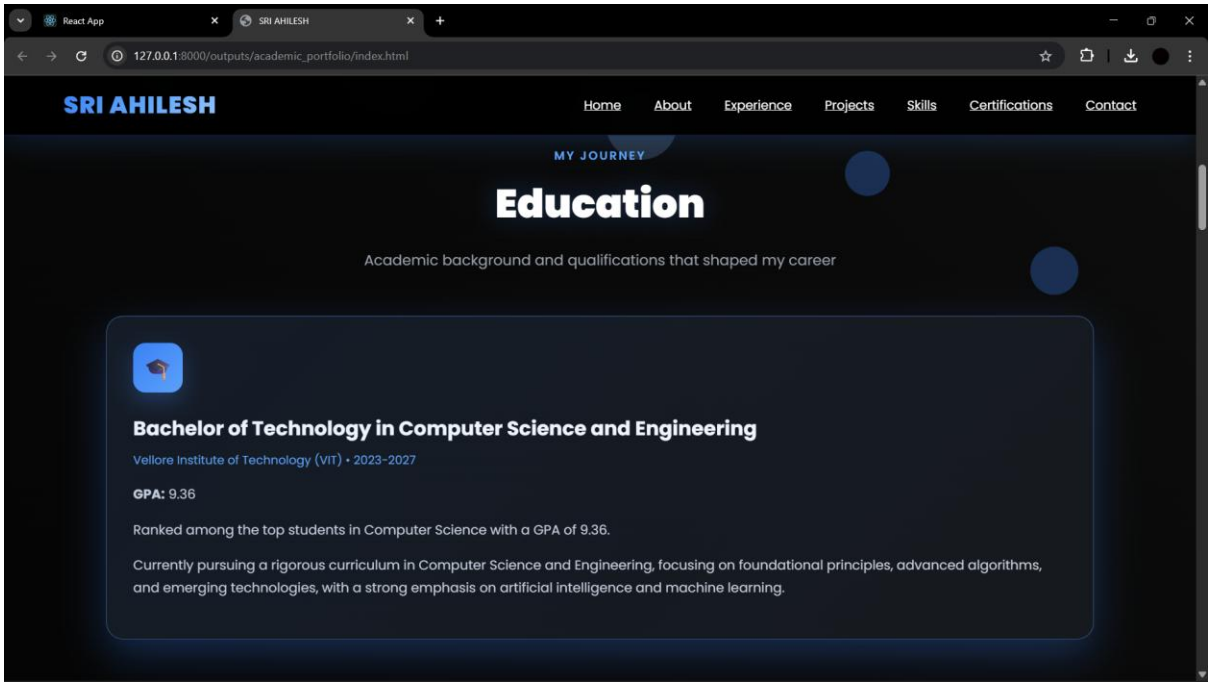
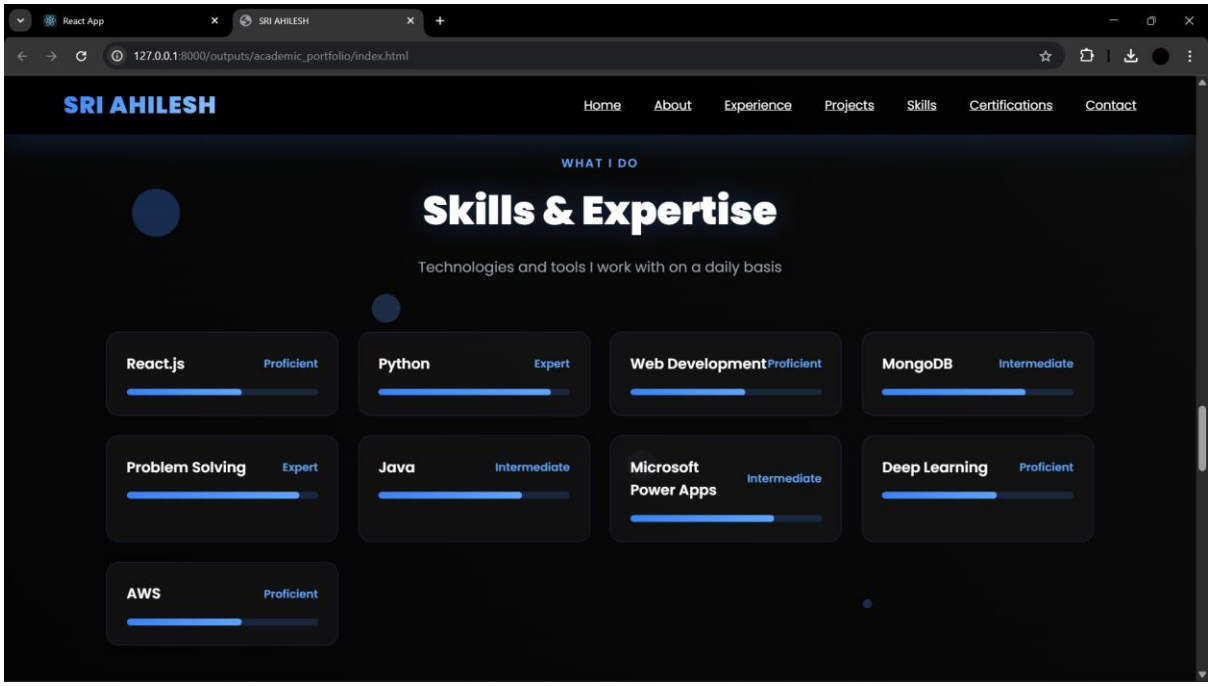
Output Page:


React App
SRI AHILESH
127.0.0.1:8000/outputs/academic_portfolio/index.html
SRI AHILESH
Home
About
Experience
Projects
Skills
Certifications
Contact
WELCOME TO MY PORTFOLIO
Hi, I'm SRI AHILESH
AI RESEARCH INNOVATOR
An ambitious Computer Science student specializing in AI/ML, dedicated to advancing computational methodologies and innovative application development. Eager to leverage technical expertise and analytical rigor for impactful research and cutting-edge solutions.
View My Work
Download Resume


React App
SRI AHILESH
127.0.0.1:8000/outputs/academic_portfolio/index.html
SRI AHILESH
Home
About
Experience
Projects
Skills
Certifications
Contact
MY JOURNEY
Education
Academic background and qualifications that shaped my career
Bachelor of Technology in Computer Science and Engineering
Vellore Institute of Technology (VIT) • 2023-2027
GPA: 9.36
Ranked among the top students in Computer Science with a GPA of 9.36.
Currently pursuing a rigorous curriculum in Computer Science and Engineering, focusing on foundational principles, advanced algorithms, and emerging technologies, with a strong emphasis on artificial intelligence and machine learning.



7. Discussion

Interpretation of Results

The results show that the Agentic AI approach is effective in converting raw, unstructured Notion data into refined professional content. The agent demonstrated reasoning ability in filling gaps, maintaining tone, and generating structured sections consistently.

Advantages

- Very low user effort
- High-quality, enriched writing
- Consistent academic/industry tone
- Automated structuring and HTML rendering

Limitations

- No long-term memory
- Dependent on correctness of Notion inputs
- Limited customization of visual themes

Observations on Agent Behavior

The agent shows strong autonomy in organizing content but relies heavily on clear schemas to avoid formatting issues. It adapts well to different portfolio purposes (academic vs. industry) but avoids hallucination only because of strict prompting.

8. Conclusion

This project successfully developed an Agentic AI-powered system that generates complete academic and industry-ready portfolios directly from a user's Notion page. By integrating the Notion API, FastAPI backend, and Gemini 2.5 Flash model, the system automates data extraction, enrichment, and HTML portfolio creation with minimal user involvement. The approach demonstrates that AI can meaningfully enhance digital identity creation by producing structured, professional, and context-aware content. Future improvements may include multi-agent workflows (writer–reviewer–designer pipeline), richer portfolio templates, vector-based memory for personalization, and expanded integrations with services like LinkedIn, GitHub, and Notion databases.

9.References (IEEE Format)

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